Overall Architecture: 6-Tier GNN Hierarchy ($\mathcal{M}_1 \to \mathcal{M}_{5B}$) & Dual-Stream Equivariant PhyloGNN on $\mathcal{G}_\mathcal{T}$ ($|V| = 295, |E_{\text{dir}}| = 588$)

1. Bidirected Graph $\mathcal{G}_\mathcal{T} = (V, E_{\text{dir}})$ (PyG Data)

- $|V| = 295$ $(V_{\text{leaf}} = \{0, \ldots, 147\},\ V_{\text{int}} = \{148, \ldots, 294\})$
- $|E_{\text{dir}}| = 588$ $\left(e_\uparrow = (v \to u,\ +1) \ \oplus\ e_\downarrow = (u \to v,\ -1)\right)$
- $\mathbf{X}_k \in \mathbb{R}^{295\times 6}$: $\left[x^{\text{obs}}_{k,v},\ \delta_{k,v},\ m^{\text{leaf}}_v,\ r^{\text{root}}_v,\ T_v,\ \tilde{d}_v\right]$
  $\left(\delta_{k,v} = x^{\text{obs}}_{k,v} - \bar{x}_k;\ x^{\text{obs}}_{k,u} = \delta_{k,u} = 0\ \forall u \in V_{\text{int}}\right)$
- $\mathbf{E} \in \mathbb{R}^{588\times 4}$: $\left[\Delta t_e,\ (\Delta t_e + 10^{-3})^{-1},\ \ln \Delta t_e,\ \text{dir}_e\right]$

2. DirectedBottomUpTreeMPNNLayer ($u = 148 \to 294$)

- **Precision Gates:** $d_L = (v_l + \Delta t_l)e^{g_{L,u}},\quad d_R = (v_r + \Delta t_r)e^{g_{R,u}}$
- **Convex Node State (Stream 2):**
  $$H_{k,u} = \frac{d_R(u)H_{k,l} + d_L(u)H_{k,r}}{d_L(u) + d_R(u)},\quad v_u = \frac{d_L(u)d_R(u)}{d_L(u) + d_R(u)}$$
- **Standardized Contrast (Stream 1):**
  $$C_{k,u} = \frac{H_{k,l} - H_{k,r}}{\sqrt{d_L(u) + d_R(u)}} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_k)\ \perp\ \mu_k$$

3A. Stream 1 (QuadraticContrastScaleHead): $\hat{\sigma}_k \perp \mu_k$

- $\hat{\sigma}_{k,\mathcal{M}_{5A}} = \gamma_{c_4}\sqrt{\sum_{u=148}^{294} w_{\sigma,u}\, C^2_{k,u}},\quad \mathbf{w}_\sigma \in \Delta^{146},\quad \gamma_{c_4} \approx c_4^{-1}$
- $\mathbf{x}_k + c\mathbf{1}_{148} \ \mapsto\ \hat{\sigma}_{k,\mathcal{M}_{5A}}$ (Translation-Invariant, df = 147)

3B. Stream 2 (EquivariantRootLocationHead): $\hat{\mu}_k + c$

- $\hat{\mu}_{k,\mathcal{M}_{5A}} = H_{k,294} + \sum_{u=148}^{293} w_{\mu,u}\,(H_{k,u} - H_{k,294})\quad (H_{294} = \hat{\mu}_{\text{GLS}})$
- $\mathcal{L}_{\text{NLL}} = \frac{1}{2}\left[\frac{(\hat{\mu}_k - \mu_k)^2}{\hat{\sigma}^2_k} + \ln \hat{\sigma}^2_k\right] + \lambda_\sigma(\hat{\sigma}_k - \sigma_k)^2 + \lambda_{\text{rel}}\left(\frac{\hat{\sigma}_k - \sigma_k}{\sigma_k}\right)^2$

4. Six-Tier GNN Architectural Ablation Hierarchy ($\mathcal{M}_1 \to \mathcal{M}_{5B}$, Test_1000)

Model	Inductive Bias/Loss	Test_1000 Metrics
$\mathcal{M}_1$ (DeepSets)	$E = \emptyset$ (Point Cloud MLP)	MSE($\hat{\mu}$) = 0.6844, $R^2(\hat{\sigma})$ = 0.5686
$\mathcal{M}_2$ (GCN)	$\Delta t_e \equiv 1$ (Unweighted Graph)	MSE($\hat{\mu}$) = 0.5859, $R^2(\hat{\sigma})$ = 0.8867
$\mathcal{M}_3$ (GINE)	$u \leftrightarrow v$ (Isotropic + 0-Leak)	MSE($\hat{\mu}$) = 0.5472, $R^2(\hat{\sigma})$ = 0.9790
$\mathcal{M}_4$ (RawMSE)	$v \to u$ Postorder + $\mathcal{L}_{\text{MSE}}$	MSE($\hat{\mu}$) = 0.7933, $R^2(\hat{\sigma})$ = 0.9965
$\mathcal{M}_{5A}$ (PhyloGNN)	Dual-Stream + $\hat{\sigma}_k^{-2}\,\mathcal{L}_{\text{NLL}}$	MSE($\hat{\mu}$) = 0.7584, $R^2(\hat{\sigma})$ = **0.9966**
$\mathcal{M}_{5B}$ (BayesMMSE)	$\mathcal{M}_{5A}$ + $\text{TN}_{[-5,5]}$ & **Grid$_{250}$**	MSE($\widehat{\mu}$) = **0.5311** (**−30.2%**)

5. AmortizedEmpiricalBayesShrinkageHead ($\mathcal{M}_{5B}$ — Efron 2010; Robbins 1956)

- **Root Noise Scale:** $\hat{s}_{0,k} = \hat{\sigma}_{5A}\sqrt{v_\rho/N} = 0.4496\,\hat{\sigma}_{5A}/\sqrt{N},\quad \alpha_k = \frac{-5-\hat{\mu}_{5A}}{\hat{s}_{0,k}},\quad \beta_k = \frac{5-\hat{\mu}_{5A}}{\hat{s}_{0,k}}$
- **1D Truncated-Normal Posterior Location Readout on $[-5,5]$ + Residual MLP:**
  $$\hat{\mu}_{k,\mathcal{M}_{5B}} = \text{clip}_{[-5,5]}\left(\hat{\mu}_{5A} + \gamma_\mu\,\hat{s}_{0,k}\frac{\phi(\alpha_k) - \phi(\beta_k)}{\Phi(\beta_k) - \Phi(\alpha_k)} + \hat{s}_{0,k}\,\delta_{\mu,\text{MLP}}(\mathbf{z}_k)\right) \implies \mathbf{-30.18\%}\ \text{MSE}(\hat{\mu})$$
- **250-Point Log-Normal Prior $\pi(\sigma_g) \times \chi^2_{147}$ Quadrature & N-Tree Cohort Updating:**
  $\hat{\sigma}_{k,\mathcal{M}_{5B}} = (1-\gamma_\sigma)\hat{\sigma}_{5A} + \gamma_\sigma \sum_{g=1}^{250} w_{k,g}\sigma_g,\quad N = 10^5 \implies |\hat{\mu}_{(N)} - \mu^*| = \mathbf{0.000418}\ (0.013\%)$

6A. Classical Gold Standard: phylolm $\mathcal{O}(n)$ 3-Point Pruning (Ho & Ané, 2014 [3])

- **Postorder 6-Register Recursion** ($v = 0 \to 294$) : $\text{denom}_u = 1 + t_u(\text{vec}11_l + \text{vec}11_r),\ \mathcal{O}(n)$ time
- **Simultaneous Readout** (147 DOF) : $\hat{\mu}_{\text{phylolm}} = \frac{P_{1,\rho}}{\text{vec}11_\rho},\quad Q_{\text{int}} = PP_\rho - \frac{P^2_{1,\rho}}{\text{vec}11_\rho},\quad \hat{\sigma}_{\text{REML}} = \sqrt{\frac{Q_{\text{int}}}{147}}$
- **Exact Isomorphism:** $\max_k |\hat{\mu}_{\text{phylolm}} - \hat{\mu}_{\text{Cholesky}}| = 3.55 \times 10^{-15};\ r(\hat{\sigma}_{\mathcal{M}_{5A}}, \hat{\sigma}_{\text{phylolm}}) = \mathbf{1.000000}$

6B. Plug-and-Play Post-BM Simulation-Based Inference (SBI) Extension [22–30]

- **Post-BM SDEs:** OU $\left(\text{d}X_t = \alpha(\theta - X_t)\text{d}t + \sigma\text{d}W_t\right)$, Bounded BM ($X_t \in [a,b]$), Lévy Jumps $\text{d}J_t$
- **Analytical Limit:** Closed-form Cholesky $\mathbf{S}^{-1}$ & PIC break down when $p(\mathbf{X} \mid \boldsymbol{\theta})$ has no closed form
- **100% Reusable GNN:** $\mathcal{G}_\mathcal{T}$ (295 nodes) + TreeMPNN + $\mathcal{M}_{5B} \ \mapsto\ \hat{\boldsymbol{\theta}}_{\text{MMSE}}(\mathbf{X}) \approx \mathbb{E}[\boldsymbol{\theta} \mid \mathbf{X}]$ (**41.2 ms**)